Note

- $(x-a)$ is a factor of $p(x)$, if $p(a)=0$ $\quad(\because x-a=0,\ x=a)$
- $(x+a)$ is a factor of $p(x)$, if $p(-a)=0$ $\quad(\because x+a=0,\ x=-a)$
- $(ax+b)$ is a factor of $p(x)$, if $p\left(-\frac{b}{a}\right)=0$ $\quad\left(\because ax+b=0,\ ax=-b,\ x=-\frac{b}{a}\right)$
- $(ax-b)$ is a factor of $p(x)$, if $p\left(\frac{b}{a}\right)=0$ $\quad\left(\because ax-b=0,\ ax=b,\ x=\frac{b}{a}\right)$
- $(x-a)(x-b)$ is a factor of $p(x)$, if $p(a)=0$ and $p(b)=0$ $\left(\because \begin{matrix} x-a=0 & \text{or} & x-b=0 \\ x=a & \text{or} & x=b \end{matrix}\right)$

Example 3.13

Show that $(x+2)$ is a factor of $x^3-4x^2-2x+20$

Solution

Let $p(x)=x^3-4x^2-2x+20$

By factor theorem, $(x+2)$ is factor of $p(x)$, *if* $p(-2)=0$

$$p(-2)=(-2)^3-4(-2)^2-2(-2)+20$$
$$=-8-4(4)+4+20$$
$$p(-2)=0$$

Therefore, $(x+2)$ is a factor of $x^3-4x^2-2x+20$

To find the zero of x+2;

put $x+2=0$

we get $x=-2$

Example 3.14

Is $(3x-2)$ a factor of $3x^3+x^2-20x+12$?

Solution

Let $p(x)=3x^3+x^2-20x+12$

By factor theorem, $(3x-2)$ is a factor, if $p\left(\frac{2}{3}\right)=0$

$$p\left(\frac{2}{3}\right)=3\left(\frac{2}{3}\right)^3+\left(\frac{2}{3}\right)^2-20\left(\frac{2}{3}\right)+12$$
$$=3\left(\frac{8}{27}\right)+\left(\frac{4}{9}\right)-20\left(\frac{2}{3}\right)+12$$
$$=\frac{8}{9}+\frac{4}{9}-\frac{120}{9}+\frac{108}{9}$$

To find the zero of $3x$–2;

put $3x-2=0$

$3x=2$

we get $x=\frac{2}{3}$

Progress Check

1. $(x+3)$ is a factor of $p(x)$, if $p(__)=0$
2. $(3-x)$ is a factor of $p(x)$, if $p(__)=0$
3. $(y-3)$ is a factor of $p(y)$, if $p(__)=0$
4. $(-x-b)$ is a factor of $p(x)$, if $p(__)=0$
5. $(-x+b)$ is a factor of $p(x)$, if $p(__)=0$

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